

Unsupervised Domain Adaptation for Adverse Weather — Project Cheat Sheet

Cityscapes (source, clear weather) → ACDC (target, fog/rain/snow/night) • SegFormer/MiT-B4 • Phase 1: baseline • Phase 2: measure the gap
• Phase 3: close the gap via self-training UDA

THE STORY (3 phases)

Phase 1 — Build the baseline. Train a segmentation model on labeled clear-weather Cityscapes. Result: **73.18% mIoU** (SegFormer/MiT-B4, full-res 2048x1024, 28 epochs, OHEM-CE+Dice loss, AdamW+poly LR, multi-scale+flip TTA). Strong on big classes (road 97%, sky 93%, car 93%), weaker on thin/rare ones (pole 48%, fence 54%) — expected pattern.

Phase 2 — Measure the domain gap. Run that same model, completely unmodified, on ACDC (fog/rain/snow/night), zero adaptation. Result: **49.02% avg mIoU** (fog 60.7 / rain 52.4 / snow 47.6 / night 35.4) — a -24.16pt (-33% relative) drop from the 73.18% source score. Severity ordering (fog mildest, night worst) matches physical intuition.

Phase 3 — Close the gap (in progress). DAFormer-style self-training UDA: an EMA teacher-student pair, both initialized from the Phase 1 checkpoint. The teacher (a slow, smoothed average of the student) generates pseudo-labels on unlabeled ACDC images; the student learns from those plus keeps training on labeled Cityscapes so it never forgets clear weather. Goal: push ACDC mIoU up from 49.02% toward 73.18% — without ever touching an ACDC label.

BUGS CAUGHT & FIXED (why the numbers are trustworthy)

- P1: Kaggle Cityscapes mirror was secretly 96x256 (not 2048x1024) → capped mIoU at 21.8%; fixed by sourcing official full-resolution data.
- P1: OpenCV-thread vs. DataLoader-worker deadlock froze training at epoch 9 → fixed with `cv2.setNumThreads(0)` + per-epoch checkpoint/resume.
- P2: Evaluation model rebuilt with the wrong architecture (DeepLabV3+/ResNet34) but loaded SegFormer weights → near-zero garbage, not a real gap; fixed with strict state-dict loading.
- P2: Stale hardcoded baseline (18.73% from a discarded draft) used instead of the true 73.18%; ACDC ground-truth mask archive was missing entirely (0 masks found) — both fixed.
- P3: LR scheduler's total_iters used `min(source,target)` batches, but the loop iterates the full (larger) source_loader → schedule would overrun, and `(1-progress)**0.9` on a negative number silently returns a Python complex number, corrupting/crashing training. Fixed: `steps_per_epoch = len(source_loader)`.
- P3: Pseudo-label confidence was discounted twice (per-pixel masking + scalar weight), risking near-zero learning signal for hard conditions like night. Simplified to match DAFormer exactly (scalar weight only).

GLOSSARY

mIoU	Mean Intersection-over-Union: per-class $(\text{predicted} \cap \text{truth}) / (\text{predicted} \cup \text{truth})$, averaged over classes. 0-100%, higher = better.
Domain gap	Performance drop when a model sees data unlike its training distribution (e.g. clear→bad weather).
UDA	Unsupervised Domain Adaptation: adapt to an unlabeled target domain using labeled source + unlabeled target data only.
Pseudo-label	A model-generated guess used as a stand-in training target where no ground truth exists.
EMA teacher-student	Student updates via backprop each step; teacher = slow exponential moving average of student weights ($\alpha=0.999$), never trained directly — stabler pseudo-labels.
Weak/strong aug	Teacher sees a clean (weak) view to generate the pseudo-label; student trains on a color-jittered (strong) view of the same pixels — forces robustness.
Confidence weighting	Pseudo-label loss scaled by the fraction of pixels the teacher is ≥ 0.968 confident about.
TTA	Test-time augmentation: average predictions over multiple views (flip/multi-scale) at inference only.
OHEM / Dice loss	OHEM: train harder on the highest-loss pixels only. Dice: loss that directly rewards shape overlap, helps class imbalance.
SegFormer/MiT-B4	Transformer segmentation model (hierarchical Mix-Transformer encoder + lightweight MLP decoder) used in all 3 phases.
DAFormer / MIC	DAFormer (Hoyer et al., CVPR'22): EMA self-training UDA recipe used here. MIC (CVPR'23): masked-patch consistency for SOTA results — sketched, not yet enabled.

LIKELY PROFESSOR Q&A

Q: Why not just collect labeled bad-weather data?

A: Pixel-level labeling is extremely expensive/slow — UDA is valuable precisely because it avoids that cost.

Q: How do you know 49.02% is a real gap, not another bug?

A: Verified checkpoint params/keys match the architecture, image/label counts matched exactly per condition, and severity ordering (fog mildest, night worst) matches physical expectation.

Q: What's novel vs. just re-running DAFormer?

A: Rigorous reproduction (bugs caught at every phase) plus testing whether weather-aware masking beats generic MIC masking — the proposal's further objective, still ahead.

30-SECOND PITCH

"We're closing the gap between how well AI street-scene understanding works in good vs. bad weather, without any labeled bad-weather data. Phase 1 built a strong clear-weather baseline (73% mIoU). Phase 2 showed that drops to ~49% in fog/rain/snow/night — a verified 24-point gap. Phase 3 recovers that gap via self-training: the model generates its own pseudo-labels for bad-weather images and learns from its confident guesses, while continued training on labeled clear-weather data keeps it from forgetting what it already knows."